

Project Summary

Project Overview

The **End-to-End Retail Performance and Behavioral Analytics** project was developed to analyze retail business data and generate actionable insights using **Python, SQL Server, and Power BI**. The project covered the complete analytics workflow, including data cleaning, database creation, SQL-based business analysis, and the development of interactive dashboards to support data-driven decision-making.

Project Objectives

- Analyze sales performance and revenue trends.
 - Understand customer behavior and purchasing patterns.
 - Evaluate product and category performance.
 - Analyze store performance and profitability.
 - Examine product return behavior and return reasons.
 - Develop interactive Power BI dashboards with key business KPIs and visualizations.
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Tools Used

- **Python (Pandas)** – Data cleaning, preprocessing, and exploratory data analysis.
 - **SQL Server** – Database creation, data management, relationship creation, and business analysis.
 - **Power BI** – Interactive dashboards and data visualization.
 - **CSV Files** – Source data.
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Project Workflow

1. Collected and explored retail datasets.
 2. Cleaned and transformed the data using Python (Pandas).
 3. Exported the cleaned datasets as CSV files.
 4. Imported the cleaned datasets into SQL Server.
 5. Created database tables, primary keys, foreign keys, and an ER diagram.
 6. Performed SQL queries to analyze sales, customers, products, stores, and returns.
 7. Connected SQL data to Power BI and created data relationships.
 8. Built **five** interactive dashboards with KPIs, charts, slicers, and filters.
 9. Generated business insights and recommendations based on the analysis.
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Project Structure

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Retail_Sales_Capstone/
├── data/
│   ├── raw/
│   │   ├── sales_data.csv
│   │   ├── customers.csv
│   │   ├── products.csv
│   │   ├── stores.csv
│   │   └── returns.csv
│   └── cleaned/
│       ├── cleaned_sales.csv
│       ├── cleaned_customers.csv
│       ├── cleaned_products.csv
│       ├── cleaned_stores.csv
│       └── cleaned_returns.csv
├── notebooks/
│   └── data_cleaning_eda.ipynb
├── sql/
│   ├── business_analysis.sql
│   └── ER_diagram.png
├── dashboard/
│   └── Retail_Sales_Report.pbix
└── docs/
    ├── final_report.pdf
    └── data_cleaning_steps.pdf
```

Author Information

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Tools Used: Python (Pandas), SQL Server, Power BI